

Voltammetry is a group of electroanalytical techniques which deals with the interplay between electricity and chemistry, namely, the measurements of electrical quantities, such as current, potential, or charge and their relationship to chemical parameters. Such use of electrical measurements for analytical purposes has found a vast range of applications, including environmental monitoring, industrial quality control, or biomedical analysis. In contrast to many chemical measurements, which involve homogeneous bulk solutions, electrochemical processes take place at the electrode-solution interface. The term voltammetry is derived from electrical parameters/units, *volt-am (pere)-metry*. Voltammetry may be defined as "the study of current voltage relation at a micro electrode, called working electrode". In other words, the technique is based upon the potential current behavior of a polarizable electrode in the solution being analyzed. According to the recommendation of IUPAC, voltammetry is the general term to be used when the current potential relationships are being investigated. The surface area of voltammetry electrode is very small (0.0001 m^2) which is helpful in providing response of current produced by species in solutions. It is also important to clarify here that when mercury (flowing through a fine capillary seems to be emerging as a droplet) is used as a small electrode in voltammetric cell, the technique has a specific name, *polarography*.

Basic principle of voltammetry

All the electrochemical techniques involve the use of electricity either as an input or output signal. The principle of voltammetry technique is to generate an input electrical signal and to measure corresponding output electrical signal. The input signals can be a preprogrammed function of time whereas the output signals may change with time due to sample under investigation. The signals are typically voltage, current and charge where charge is the integration of current with time. The input electrical signals usually cause the electrochemical reactions at working and counter electrodes. Simply, a measured small potential is established across a pair of electrodes, one of which is a non-polarizable reference electrode and the other a polarizable inert electrode. The current which flows depends upon composition of the solution.

Voltammogram

A plot of current as a function of applied potential is called a *Voltammogram*. It is the electrochemical equivalent of a spectrum in spectroscopy, providing quantitative and qualitative information about the species involved in the oxidation or reduction reaction



where O and R are the oxidized and reduced forms, respectively, of the redox couple. Such a reaction will occur in a potential region that makes the electron transfer thermodynamically or kinetically favorable. For systems controlled by the laws of thermodynamics, the potential of the electrode can be used to establish the concentration of the electroactive species at the surface $[\text{Co}(0,t)$ and $\text{Cr}(0,t)]$ according to the Nernst equation. On the negative side of E° , the oxidized form thus tends to be reduced, and the forward reaction (i.e., reduction) is more favorable. The current resulting from a change in oxidation state of the electroactive species is

termed the *faradaic current* because it obeys Faraday's law (i.e., the reaction of 1 mol of substance involves a change of $n \times 96,487\text{C}$). The faradaic current is a direct measure of the rate of the redox reaction.

The pathway of the electrode reaction can be quite complicated, and takes place in a sequence that involves several steps. The rate of such reactions is determined by the slowest step in the sequence. Simple reactions involve only mass transport of the electroactive species to the electrode surface, electron transfer across the interface, and transport of the product back to the bulk solution. More complex reactions include additional chemical and surface processes that either precede or follow the actual electron transfer. The net rate of the reaction, and hence the measured current, may be limited by either mass transport of the reactant or the rate of electron transfer. The more sluggish process will be the rate-determining step. Whether a given reaction is controlled by mass transport or electron transfer is usually determined by the type of compound being measured and by various experimental conditions (electrode material, media, operating potential, mode of mass transport, time scale, etc.).

Cyclic voltammetry

One of the more complex electrochemical techniques, cyclic voltammetry is very frequently used because it offers a wealth of experimental information and insights into both the kinetic and thermodynamic details of many chemical systems. Cyclic voltammetry consists of scanning linearly the potential of a stationary working electrode (in an unstirred solution), using a triangular potential waveform (Figure 10.2). Depending on the information sought, single or multiple cycles can be used. During the potential sweep, the potentiostat measures the current resulting from the applied potential. The resulting current-potential plot is termed a cyclic voltammogram. The cyclic voltammogram is a complicated, time-dependent function of a large number of physical and chemical parameters.

Cyclic voltammetry is often the first experiment performed in an electroanalytical study. In particular, it offers a rapid location of redox potentials of the electroactive species, and convenient evaluation of the effect of media on the redox process. Figure 7.9 illustrates that only the oxidized form O is present initially. Thus, a negative-going potential scan is chosen for the first half-cycle, starting from a value where no reduction occurs.

The characteristic peaks in the cycle voltammogram are caused by the formation of the diffusion layer near the electrode surface. These can be best understood by carefully examining the concentration-distance profiles during the potential scan. The most common experimental configuration for recording cyclic voltammogram consists of an electrochemical cell that has three electrodes, i.e., counter or auxiliary electrode (C), reference electrode (R), and working electrode (W), all immersed in a liquid and connected to a potentiostat. The potentiostat allows the potential difference between the reference and working electrode to be controlled with minimal interference from IR (ohmic) drop. In this configuration, the current flowing through the reference electrode also can be minimized thereby avoiding polarization of the reference electrode and hence keeping the applied potential distribution between the working and reference electrode stable. Positioning the reference electrode close to the working electrode further helps to minimize the IR

drop between the reference and working electrode due to the resistivity of the solution phase.

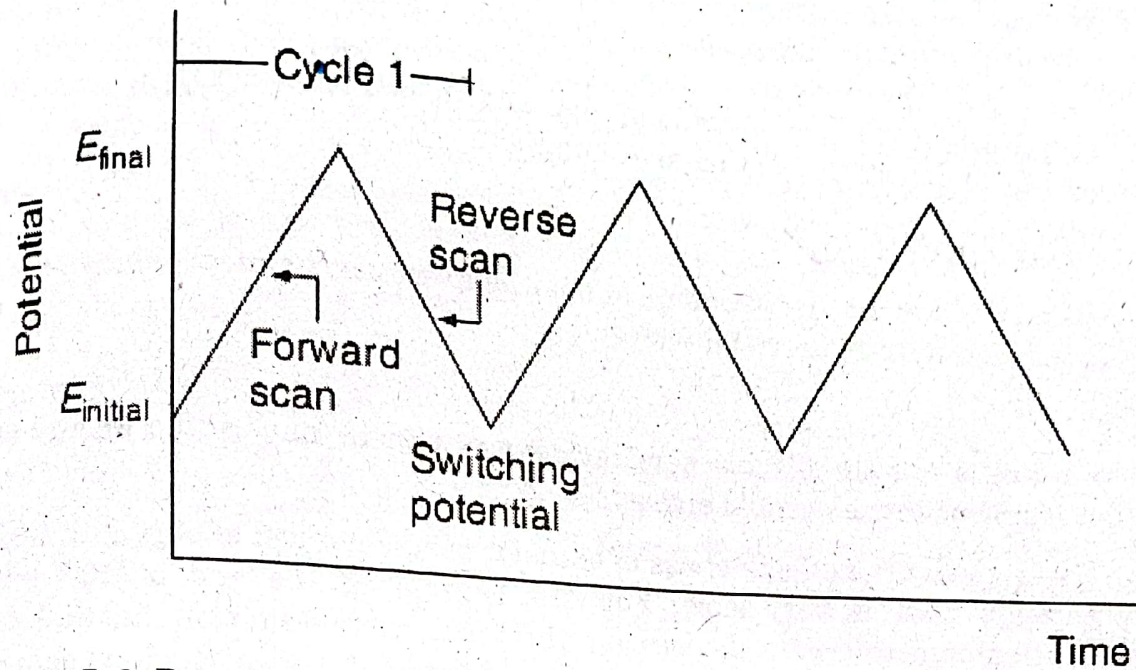


Figure 7. 9. Potential-time excitation signal in a cyclic voltammetric experiment.

Reference electrode: Commonly used are aqueous Ag/AgCl or calomel half cells, which can be obtained commercially or easily prepared in the laboratory.

Counter electrode: Non-reactive high surface area electrode, commonly platinum gauze.

Working electrode: Most commonly used are inlaid disc electrodes (Pt, Au, graphite, glassy carbon, etc.) of well-defined area. Other geometries may be employed beneficially in appropriate circumstances (dropping or hanging mercury hemisphere, cylinder, band, arrays, or grid electrodes).

Data Interpretation

The cyclic voltammogram is characterized by several important parameters. Four of these observables, the two peak currents and two peak potentials, provide the basis for the diagnostics developed by Nicholson and Shain for analyzing the cyclic voltammetric response.

Stripping voltammetry

Stripping voltammetry is an extremely sensitive electrochemical technique for measuring trace metals. Its remarkable sensitivity is attributed to the combination of an effective pre-concentration step with advanced measurement procedures that generate an extremely favorable signal background ratio. Since the metals are pre-concentrated into the electrode by factors of 100–1000, detection limits are lowered by two to three orders of magnitude compared to solution-phase voltammetric measurements. Hence, four to six metals can be measured simultaneously in various matrices at concentration levels down to 10^{-10} M, utilizing relatively inexpensive instrumentation. The ability to obtain such low detection limits strongly depends on the degree to which contamination can be minimized. Expertise in ultra-trace chemistry is required.

Essentially, stripping analysis is a two-step technique. The first, or *deposition step*, involves the electrolytic deposition of a small portion of the metal ions in solution into the mercury electrode to pre-concentrate the metals. This is followed by the *stripping step* (the measurement step), which involves the dissolution (stripping) of the deposit. Different versions of stripping analysis can be employed, depending on the nature of the deposition and measurement steps.